

Principle of operation

Induction motor



3φ = each phase
3φ Supply 120° apart



→ short circuited.

magnetic field feels rotating As it is pulsating, the rotor conductor cuts the flux lines will cut this.

Rotor conductors made on induced emf. Not

By ~~the~~ Lenz's Law, to avoid the cutting of ~~the~~ flux lines, the rotor should also move in the same way.

It will ~~never~~ only rotate near to Synchronous speed.

To find the direction of rotation, Direction of rotation is the same here.

Rotating Speed of stator magnetic field

$$N_s = \frac{120f}{p} \rightarrow \text{freq. of supply Voltage}$$

Speed of Rotor = N

Slip is the difference in stator field & Rotor speed denoted by S .

Load ↑ I ↓ Cutting ↑

Till then you give torque.

Slip represents fraction symbol or

$$S = \frac{N_s - N}{N_s} \times 100\%$$

The frequency of Synchronous Speed depends on $N_s = \frac{120f}{p}$ ①

Induced I frequency depends on slip.

Frequency of rotor Current or emf

Rotor emf frequency $f' = \frac{\text{Relative Speed in rpm}}{120/p}$

$$= \frac{N_s - N}{120/p} \quad \left| \begin{array}{l} \text{Since} \\ S = \frac{N_s - N}{N_s} \end{array} \right.$$

$$S N_s = N_s - N$$

$$S \times \frac{120f}{p} = N_s - N$$

$$f' = \frac{N_s - N}{120/p} = \frac{S \times 120f}{p} \times \frac{p}{120/p} = S f$$

$$\underline{\underline{f' = S f}}$$

Rotor frequency = S .

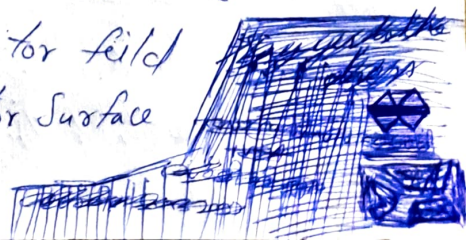
Speed of stator field w.r.t to stator = N_s

Speed of rotor w.r.t stator = N

Speed of stator field w.r.t rotor structure = $N_s - N$

Speed of rotor field w.r.t stator surface

& is



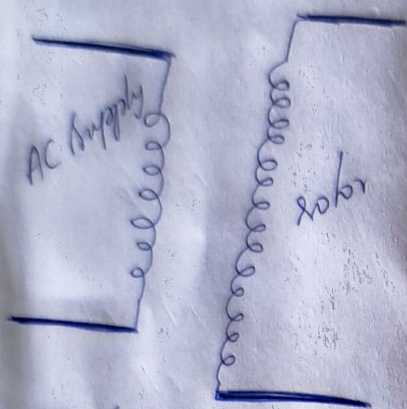
$$= N + sNs$$

$$= Ns(1-s) + sNs$$

$$= \underline{Ns}$$

If Line Voltage
given in exam
divide by $\sqrt{3}$ to
make it phase
Voltage

Winding Dia of Induction motor



Behaves as a transformer at
stand still (B4 the rotor rotates)
as the slip value = 1

Rotor Emf, E_2

$$E_2 = E_1 \frac{N_2}{N_1}$$

where N_2 = rotor winding (no. of turns per phase)

at stand still, $N = 0$

$$s = \frac{N_s - 0}{N_s}$$

$$\underline{s = 1}$$

N_1 = stator winding turns

E_1 = applied Voltage Per phase

E_2 = Induced rotor Voltage/phase

Rotor Current & powerfactor

Let R_2 = rotor Resistance per phase

X_2 = Rotor Inductance per phase

E_2 - Rotor Induced emf / phase ②

At stand still Condition $s = 1$

$$f' = sf$$

Rotor reactance / phase $\underline{f' = f}$

$$X_2 = 2\pi f l$$

Rotor winding Impedance / phase

$$Z_2 = \sqrt{R_2^2 + X_2^2}$$

Rotor Current / per phase = I_2

$$I_2 = \frac{E_2}{Z_2} = \frac{E_2}{\sqrt{R_2^2 + X_2^2}}$$

Pf of rotor Current $\cos \phi_2$

$$= \frac{R_2}{\sqrt{R_2^2 + X_2^2}}$$

At slip s = Running Condition

Induced emf per phase = sE_2

Reactance / phase = sX_2 ($2\pi f' L$)

$$= 2\pi s f L$$

$$= sX_2$$

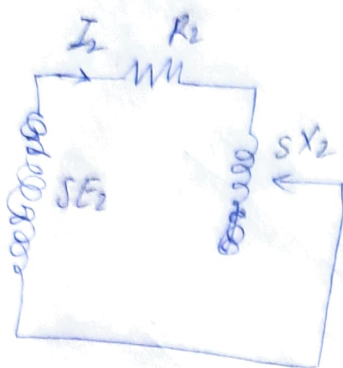
Rotor winding Impedance / phase

$$= \sqrt{R^2 + (sX_2)^2}$$

$$I_2 = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}}$$

$$\cos \phi_2 = \frac{R_2}{\sqrt{R_2^2 + (sX_2)^2}}$$

Equivalent Ckt of a Rotor ckt

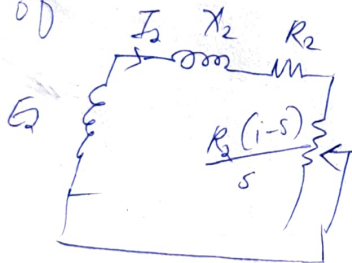


$$I_2 = \frac{SE_2}{\sqrt{R_2^2 + (sX_2)^2}}$$

$$I_2 = \frac{E_2}{\sqrt{\left(\frac{R_2}{s}\right)^2 + X_2^2}} = \frac{R_2 + R_2(1-s)}{s}$$

$$I_2 = \frac{E_2}{\sqrt{\left(\frac{R_2}{s}\right)^2 + X_2^2}} \quad \omega \frac{R_2}{s} = f$$

✓
This resistance consumes
cheyenne power as mechanical
power of the motor



Rotor emf/phase $E_2 \propto \phi$
(flux/pole)

$$T \propto I_2 E_2 \cos \phi_2$$

$$T = k \cdot I_2 E_2 \cos \phi_2$$

write values of I_2 and $E_2 \cos \phi_2$
with slip including this.

At running Condition,

$$I_2 = \frac{SE_2}{\sqrt{R_2^2 + (sX_2)^2}}$$

$$\cos \phi_2 = \frac{R_2}{\sqrt{R_2^2 + (sX_2)^2}}$$

$$T = \frac{k E_2 \times SE_2 \times R_2}{R_2^2 + (sX_2)^2}$$

$$= \frac{k \cdot s \cdot E_2^2 R_2}{R_2^2 + (sX_2)^2}$$

Impedance

At starting $s=1$

$$T_{st} = \frac{k \cdot R_2 \cdot E_2^2}{R_2^2 + (sX_2)^2}$$

Condition for maximum ~~starting~~ ^{running} Torque

Torque developed by rotor is
directly proportional to the rotor

Current I_2

stator flux per pole, ϕ

Power factor of the rotor circuit

$\cos \phi_2$

i.e., $T \propto I_2 \phi \cos \phi_2$

$$T = \frac{k R_2 S E_2^2}{R_2^2 + (S X_2)^2}$$

T for fixed supply will be maximum when $\frac{S R_2}{R_2^2 + (S X_2)^2}$ is maximum

